Sunbeom Kwon

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EDUCATION

University of Illinois Urbana-Champaign, Champaign, IL 2026
Ph.D. in Quantitative Psychology

University of Illinois Urbana-Champaign, Champaign, IL 2024
M.S. in Statistics

Sungkyunkwan University, Seoul 2020
M.A. in Psychology

Sungkyunkwan University, Seoul 2015
B.A. in Psychology and Statistics

EMPLOYMENT

University of Washington
Postdoctoral Scholar September 2026 - Present

ETS
Ida Lawrence Research Summer Intern June 2024 - July 2024

BSCHR, Seoul
Psychometric Researcher August 2020 - February 2021

PUBLICATIONS

Accepted:

1. **Kwon, S.** & Zhang, S. (in press). A Copula-Based Joint Model for Speed and Accuracy. *Psychometrika*
2. Jiang, H., **Kwon, S.**, Luo, J., Xiao, Z., Zhang, S. (2026). Can We Trust Item Response Theory for AI Evaluation? *Advances in Neural Information Processing Systems (NeurIPS)*
3. Zhang, S., He, Q., & **Kwon, S.** (2026). Digital Module 41: Process Data. *Educational Measurement: Issues and Practice*
4. Choi, J., Zhang, B., Shi, D., **Kwon, S.**, & Alexander, L. (2026). You Must Parcel Carefully If You Have To! Comparing Eight Item Parceling Strategies with the Item-level Model for Bifactor Predictive Models. *Psychological Methods*
5. **Kwon, S.**, McCaffrey, D., Jewsbury, P., & Casabianca, J. (2025). An Empirical Bayesian Approach for Testing the Fairness of Automated Scoring. *Journal of Educational and Behavioral Statistics*
6. **Kwon, S.**, & Zhang, S. (2025). Explaining Performance Gaps with Problem-Solving Process Data via Latent Class Mediation Analysis. *Psychometrika*
7. **Kwon, S.**, Zhang, S., Köhn, F., & Zhang, B. (2025). MCMC Stopping Rules in Latent Variable Modelling. *British Journal of Mathematical and Statistical Psychology*

8. Choi, J., **Kwon, S.**, & Zhang, B. (2025). A Structural After Measurement Approach to Bifactor Predictive Models. *Structural Equation Modeling: A Multidisciplinary Journal*

Under Review/Revision:

1. **Kwon, S.**, Wang, C., & Xu, G. DECAF: Disentangling Capability and Translation Artifacts for Multilingual LLM Evaluation. *International Conference on Learning Representations (ICLR)*

In Preparation:

1. **Kwon, S.**, Zhang, S. Predicting Binary Risk Outcomes via Copula-Based SEM.
2. **Kwon, S.**, McCaffrey, D., Jewsbury, P., & Casabianca, J. Multivariate Empirical Bayes for Item-Level Fairness Decisions in Automated Scoring.
3. **Kwon, S.**, Zhang, B., & Zhang, S. A Neural Network Approach to the Estimation of High-Dimensional Thurstonian Item Response Model.

PRESENTATIONS

Invited Talks:

University of Washington:

1. **Kwon, S.** (2026, October). Copula-Based Latent Variable Modeling for Asymmetric Dependence. *Invited presentation at the Center for Statistics and the Social Sciences Seminar.*

Leading Conference Presentations:

1. **Kwon, S.**, Zhang, S. (2026, April). A Copula-based Joint Model for Item Responses and Response Times. *Poster presented at the National Council on Measurement in Education (NCME) Annual Meeting, Los Angeles, CA.*
2. **Kwon, S.**, McCaffrey, D., Jewsbury, P., & Casabianca, J. (2026, April). Multivariate Empirical Bayes for Item-Level Fairness Decisions in Automated Scoring. *Paper presented at the National Council on Measurement in Education (NCME) Annual Meeting, Los Angeles, CA.*
3. **Kwon, S.**, Zhang, S. (2025, July). A Copula-based Joint Model for Item Responses and Response Times. *Paper presented at the International Meeting of the Psychometric Society (IMPS), Minneapolis, MN.*
4. **Kwon, S.**, Zhang, S. (2025, April). Explaining Performance Gaps with Problem-Solving Process Data via Latent Class Mediation Analysis. *Poster presented at the National Council on Measurement in Education (NCME) Annual Meeting, Denver, CO.*
5. **Kwon, S.**, McCaffrey, D., Jewsbury, P., & Casabianca, J. (2025, April). An Empirical Bayesian Approach for Testing the Fairness of Automated Scoring. *Paper presented at the National Council on Measurement in Education (NCME) Annual Meeting, Denver, CO.*
6. **Kwon, S.**, Zhang, S. (2024, July). Explaining Performance Gaps with Problem-Solving Process Data via Latent Class Mediation Analysis. *Paper presented at the International Meeting of the Psychometric Society (IMPS), Prague, Czech.*
7. **Kwon, S.**, Zhang, S. (2023, November). Explaining Performance Gaps with Problem-Solving Process Data via Latent Class Mediation Analysis. *Paper presented at the Ideas in Testing Research Seminar, Chicago, IL.*
8. **Kwon, S.**, Zhang, S., & Köhn, F. (2023, March). MCMC Convergence Diagnostics in DINA Model and Bifactor IRT Model. *Poster presented at the National Council on Measurement in Education (NCME) Annual Meeting, Chicago, IL.*

9. **Kwon, S.**, Jahng, S. (2020, October). Effects of Factor Scaling Constraints on Model Estimation in Bayesian Confirmatory Factor Analysis. *Paper presented at the Meeting of the Korean Society of Psychological Measurement and Assessment, Seoul, Korea.*
10. **Kwon, S.**, Jahng, S. (2019, March). Autoregressive Cross-lagged Model with Phantom Variables for Panel Data with a Missing Time Point. *Poster presented at the International Convention of Psychological Science, Paris, France.*
11. **Kwon, S.** (2017, November). Testing Autoregressive Cross-lagged Effects between Academic Achievement and Achievement Values Using Phantom Variables. *Poster presented at the Meeting of the Korean Society of Psychological Measurement and Assessment, Seoul, Korea.*

Contributed Conference Presentations:

1. Choi, J., **Kwon, S.,** & Zhang, B. (2024, April). An Investigation of the Structural After Measurement Approach for Bifactor Models. *Poster presented at the Society for Industrial and Organizational Psychology (SIOP) Annual Conference, Chicago, IL.*

TEACHING

University of Illinois Urbana-Champaign
Lab Instructor and Teaching Assistant
Courses:

- PSYC 490: Measurement and Test Development Lab — Fall 2022, 2023
- PSYC 301: Psychological Statistics — Spring 2025, 2024, 2023
- PSYC 235: Introduction to Statistics — Fall 2024

Sungkyunkwan University
Teaching Assistant
Courses:

- Psychological Testing and Measurement — Spring 2018
- Statistical Analysis in Psychology — Fall 2018
- Introduction to Psychology — Spring 2019

MENTORSHIP

University of Illinois Urbana-Champaign

Undergraduate Research Experience in Statistics (URES) Mentor — Spring 2026

SOFTWARE

The **hmcdm** R package — Creator / Maintainer

- A package for fitting hidden Markov cognitive diagnosis models on longitudinal data for learning.
- `https://github.com/tmsalab/hmcdm`
- `https://cran.r-project.org/web/packages/hmcdm/index.html`

SERVICE

Professional Committee:

- National Council on Measurement in Education
  Dissertation Award Committee 2023

Ad-hoc Manuscript Reviewer:

- *Behavioral Research Methods*
- *Behaviormetrika*
- *Educational Measurement: Issues and Practice*
- *Journal of Educational and Behavioral Statistics*
- *Multivariate Behavioral Research*
- *Psychometrika*
- *Statistics and Its Interface*

AWARDS & HONORS

External:

- Psychometric Society Travel Award, IMPS 2025.

University of Illinois Urbana-Champaign:

- Jeffrey Tanaka Memorial Award, 2026.
- Graduate College Conference Presentation Award, 2025.
- University Block Grant Fellowship, 2024.
- Psychology Grad Fellowship, 2025, 2026.
- Hobson Fellowship 2022, 2024.
- List of Teachers Ranked as Excellent by Students, Center for Innovation in Teaching & Learning, 2023.